

SUDAKSHINA DEB

Senior Scientist — Discovery Biology & Downstream Protein Sciences

Bengaluru, India | +91-7005314758 | sudakshinadeb97@gmail.com | linkedin.com/in/sudakshina-deb

PROFESSIONAL PROFILE

Results-driven **Senior Scientist with ~4 years of biopharmaceutical industry experience** in Discovery Biology and Downstream Protein Sciences at **Syngene International Ltd.** Proven track record leading end-to-end purification of complex biologics—including monoclonal antibodies (mAbs), bispecific antibodies (bsAbs), and His-tagged recombinant proteins—utilizing ÄKTA chromatography systems (Protein A/G, Ni-NTA, IEX, HIC, SEC) and Tangential Flow Filtration (TFF). Highly proficient in analytical release testing and quality characterization (SEC-HPLC, SDS-PAGE, Western blot, LC-MS/MS, UV-Vis, Endosafe kinetic LAL endotoxin assays) under strict GLP/ALCOA++ data integrity compliance. Seamlessly integrates wet-lab bioprocess rigor with computational genomics and bioinformatics workflows (Python, R, QIIME2, AlphaFold2). Conferred the **Syngene SPOT Award (Feb 2023)** for rapid troubleshooting and high-recovery protein delivery for global biopharma clients.

CORE TECHNICAL COMPETENCIES

- **Downstream Bioprocess & Chromatography:** ÄKTA Pure, ÄKTA Avant, UNICORN software; Affinity (Protein A/G, Ni-NTA His-tag), Ion Exchange (IEX/CEX/AEX), Hydrophobic Interaction (HIC), Size Exclusion (SEC/GFC); Desalting, Buffer Exchange, Batch Binding, Dialysis, Ultrafiltration/Diafiltration (UF/DF), Tangential Flow Filtration (TFF, Pellicon cassettes).
- **Bioanalytical Characterization & Quality Release:** SEC-HPLC (monomer purity & aggregate profiling), SDS-PAGE (reducing/non-reducing), Western Blotting, LC-MS/MS, UV-Vis Spectrophotometry, Endosafe PTS/MCS (kinetic LAL endotoxin testing/removal), Flow Cytometry, Protein Impurity Characterization.
- **Molecular Biology & Laboratory Operations:** Recombinant protein expression, PCR, Primer Design, Sanger Sequencing, Gene Expression Profiling, Aseptic Mammalian Cell Culture, Media Prep, Passaging, In Vivo Animal Handling (rodent surgical procedures, model organisms).
- **Computational Biology & Data Science:** Python (Pandas, Biopython), R (tidyverse), Linux/Bash shell scripting, NGS Genomic Data Analysis, QIIME2 (DADA2 pipeline), AlphaFold2 (ColabFold), In Silico Antibody CDR Engineering, Biological Databases (SILVA, NCBI BLAST, STRING, ExPASy).
- **Quality & Regulatory Governance:** ALCOA++ Data Integrity Standards, Standard Operating Procedures (SOPs), Batch Manufacturing Records (BMRs), GLP/GMP Laboratory Compliance, Cross-Functional Client Milestone Delivery.

PROFESSIONAL INDUSTRY EXPERIENCE

SYNGENE INTERNATIONAL LIMITED | Bengaluru, India

Senior Scientist – Discovery Biology | May 2022 – Present

- **Biotherapeutic Downstream Purification:** Lead multi-step preparative chromatography purification workflows for 40+ recombinant therapeutic projects, including mono- and bi-specific antibodies and His-tagged proteins using ÄKTA Pure and ÄKTA Avant systems.
- **Chromatography Method Development:** Developed, scaled, and standardized chromatographic purification protocols across Protein A/G, Ni-NTA, CEX, AEX, HIC, and SEC, consistently achieving >95% monomer purity and quantitative recovery yields.
- **Bioprocess & TFF Filtration:** Formulated and optimized Tangential Flow Filtration (TFF) and ultrafiltration/diafiltration (UF/DF) operational parameters, improving cycle turnaround by 25% while safeguarding product stability.
- **Analytical Release Testing:** Conducted routine release testing and quality characterization via SEC-HPLC aggregate profiling, reducing and non-reducing SDS-PAGE, Western blot validation, UV-Vis, and Endosafe kinetic LAL endotoxin quantification.
- **Cross-Functional Client Delivery:** Collaborated with cross-functional Discovery Biology and analytical teams to optimize protocols, troubleshoot challenging aggregation barriers, and ensure timely delivery of complex proteins for global biopharma clients.
- **Syngene SPOT Award (Feb 2023):** Recognized with departmental SPOT Award for producing and delivering highly challenging proteins within compressed timelines, earning direct client commendation.

RAJIV GANDHI CENTRE FOR BIOTECHNOLOGY (RGCB) | Trivandrum, India

Research Trainee – Human Molecular Genetics Laboratory | Oct 2021 – Apr 2022

- **Genetics & Molecular Workflows:** Investigated molecular and statistical genetic approaches, genotyping workflows, and candidate gene association studies in neurodevelopmental disorders.
- **Experimental Execution:** Performed primer design, PCR optimization, agarose gel electrophoresis, RNA/DNA isolation, cell culture, and Sanger sequencing analyses.
- **Computational Scripting & Publication:** Implemented bash and Python scripts for NGS data analysis and sequence alignment; synthesized findings into a review manuscript analyzing de novo genomic variations published in a peer-reviewed journal.

APPLIED BIOPHARMA & COMPUTATIONAL PROJECTS

- **Computational Antibody Design & Developability (IBAB Hackathon | Dec 2025):** Redesigned Keytruda (anti-PD-1) using CDR engineering and germline-based framework optimization. Modeled 3D tertiary structures via AlphaFold2 (ColabFold) and applied rational in silico mutagenesis to optimize binding affinity, conformational stability, and biophysical developability.
- **Oral Microbiome Biomarker Discovery for Early Cancer Detection (Jan 2026):** Investigated 16S rRNA sequencing datasets using QIIME2 (DADA2) for quality control, denoising, and ASV generation; performed alpha/beta diversity and taxonomic profiling (SILVA); developed machine learning classification models (logistic regression, decision trees) in Python/R to identify non-invasive diagnostic biomarkers.

EDUCATION & CREDENTIALS

Post Graduate Diploma in Bioinformatics & Genomics (Data Science) | Bversity (Jul 2025 – Present) — **Advanced Specialization**
Master of Science (M.Sc.) in Zoology (Honours) | North-Eastern Hill University (NEHU), Shillong (2018 – 2020) — **CGPA: 5.5 / 6.0 scale**

Bachelor of Science (B.Sc.) in Zoology (Honours) | St. Edmund's College, NEHU, Shillong (2015 – 2018) — **81.5% — First Rank Holder (Prof. D.C. Dhar Memorial Award)**

PUBLICATIONS, HONORS & CERTIFICATIONS

- **Peer-Reviewed Publication:** Deb, S. (2025). The mosaic genome: De novo variations driving neurodevelopment in autism spectrum disorder, intellectual disability, and epilepsy. *IP Indian Journal of Neurosciences*, 11(3), 133–143. doi:10.18231/j.ijn.11927.1759983172
- **Syngene SPOT Award (Feb 2023):** Conferred by Dept. of Discovery Biology for exemplary technical troubleshooting and delivering challenging biotherapeutic targets under tight client timelines.
- **Academic Honors & Awards:** Prof. D.C. Dhar Memorial Award (First Rank Holder in B.Sc. Zoology); Certificate of Academic Excellence (3rd Rank in NEHU University Merit List); Rajendra Kumar Sunheri Devi Charitable Endowment Book Grant (Top 2% of batch).
- **Specialized Certifications:** Statistical Analysis using SPSS (GISS, 2021); Molecular & Biochemistry Techniques Training (IIT Kharagpur Springfest, 2020); Neuroscience: Genetics & Development (EPFL - edX).